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Review article

The effect of cushioning materials on musculoskeletal discomfort and fatigue during prolonged standing at work: A systematic review

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ABSTRACT

This systematic review updates the current state of evidence on the effectiveness of softer flooring and cushioned shoe insoles on reducing musculoskeletal discomfort amongst workers who are required to stand for prolonged periods to work and the impact of factors such as age and gender on the outcomes. A systematic search identified 10 unique studies that met the eligibility criteria. The heterogeneity of study designs impacted on the strength of evidence. A moderate level of evidence was found in support of using cushioned materials in reducing discomfort/fatigue among standing workers. A limited level of evidence exists in favour of using insoles over anti-fatigue mats. Insufficient information exists for the impact of gender or age. Larger, good quality prospective intervention trials based in real workplaces that consider the impact of psychosocial and organisational factors on musculoskeletal discomfort whilst standing at work are required to inform industry recommendations.

1. Introduction

Despite significant changes in the way work is done since the transition from mechanisation to automation in the 1950's (Straker and Mathiassen, 2009), standing jobs remain prevalent across many industries around the world. According to the 2008 Australian National Hazard Exposure Worker Surveillance (NHEWS) survey, 31% of workers in the manufacturing, 33% in retail and 39% in the hospitality industries in Australia reported standing in one place often or all of the time (Australian Safety and Compensation Council, 2009). Prolonged standing has also been reported amongst automotive engine manufacturing plant employees in the Northern US (Gell et al., 2011) and amongst process line workers at a metal stamping company in Malaysia (Halim et al., 2013).

Musculoskeletal pain and discomfort affecting the lower back and lower extremities, leg swelling and fatigue (both local and general) has been frequently reported by those engaged in prolonged standing activities (Garcia et al., 2015; Orlando and King, 2004; Redfern and Cham, 2000; Zander et al., 2004). An increase in the risk of plantar fasciitis was found among full-time employees of a US based assembly plant (Werner et al., 2010). In their review of literature on the health effects of prolonged standing amongst workers in various occupations in Europe, Northern and Southern America, Asia and Australia, Waters and Dick (2015) found a positive association between standing still and physical fatigue, discomfort and pain.

Several physiological mechanisms have been proposed to explain the development of these adverse health effects. Musculoskeletal discomfort has been thought to be the result of inflammation caused by pooling of blood in the lower limbs secondary to reduced circulation and venous return and gravity assisted blood backflow (Hughes et al., 2011; McCulloch, 2002). Further, standing induced static muscle contraction may lead to muscle fatigue (Messing et al., 2008) and discomfort (King, 2002; Zander et al., 2004) and may trigger the development of MSDs (Garcia et al., 2015). King (2002) and Madeleine et al. (1998) suggested that the loaded muscles accumulate metabolites and become hypersensitive, further contributing to the risk of MSDs. Lack of joint movement and continuous tissue compression during prolonged standing can contribute to cartilage degeneration and rheumatic diseases over time (Canadian Centre for Occupational Health and Safety, 2008).

In contrast, phasic muscle activity helps to promote good venous return (Brantingham et al., 1970). Standing on softer surfaces has been suggested to create subtle muscular movement, reducing musculoskeletal strain, improving blood flow and decreasing discomfort and fatigue (King, 2002; Orlando and King, 2004). The fundamental principle behind using anti-fatigue mats and insoles is to provide a supportive interface between the floor and the feet and to optimise body weight distribution (White, 2002). In addition, some shoe inserts, particularly customised orthotics, provide biomechanical realignment, reduce shearing forces and alter sensorimotor control (Hatton et al., 2015).

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Material stiffness and shock absorption are considered to be the most significant determinants of discomfort during standing (Wiggermann, 2011).

There is agreement that decreasing standing time through job redesign (primary prevention) is the most effective method of reducing the risks of prolonged standing (Canadian Centre for Occupational Health and Safety, 2008; Jefferson, 2013; Ngomo et al., 2008; Van Dieen and Oude Vrielink, 1998). Unfortunately, some workers remain constrained to a standing working posture with limited opportunities to walk around or sit down, due to suboptimal job design, employment conditions, or cultural norms. For example, operating theatre clinical protocols prevent sitting down in order to protect the sterile field (Hughes et al., 2011). In places such as North America, Asia and Australia, checkout operators usually stand when serving customers, because standing is perceived to deliver better customer service and higher productivity, whereas in parts of Europe and South America this task is usually performed whilst sitting (Konz and Rys, 2002). Standing may also offer more efficiency with tasks that require heavy lifting (Messing et al., 2015).

Attempts have been made to manage standing related risk factors by using secondary level interventions including anti-fatigue mats, cushioned shoe inserts (Aghazadeh et al., 2015; King, 2002; Redfern, 1995) and replacing flooring with softer materials (Wahlstrom et al., 2012) such as rubber, carpet, vinyl or wood (Canadian Centre for Occupational Health and Safety, 2015). A number of industry bodies, among them the Association for Peri-Operative Registered Nurses (AORN) (Hughes et al., 2011), the Canadian Centre for Occupational Health and Safety (Canadian Centre for Occupational Health and Safety, 2008), Health and Safety Executive in the UK (Health and Safety Executive, 2012) and the Occupational Health Clinics for Ontario Workers (OHCOW) (Occupational Health Clinics for Ontario Workers Inc., 2012), have developed guidelines which recommend these methods.

The effects of various types of cushioning materials have been investigated using subjective reporting of fatigue and discomfort and objective measures including changes in local muscle activity, weight shifting, and changes in leg volume and local skin temperature. Previous reviews on this subject provided mixed results and focused largely on a general overview of the health outcomes of prolonged standing at work using a wide range of interventions. The methodological and reporting quality of these reviews was limited (Halim & Omar, 2011; Mohd Noor et al., 2013; Redfern and Cham, 2000; Waters and Dick, 2015). Redfern and Cham (2000) reviewed 11 studies published in the period between 1988 and 1999 and concluded that overall, softer floors were associated with decreased reports of lower back and leg pain. Material properties characterised by greater elasticity, stiffness and thickness contributed to greater comfort by improved absorption and transmission of forces. However, due to significant methodological differences between the studies, the results of objective outcomes such as leg volume, weight shifting, electromyography (EMG) findings and skin temperature, were mixed. The researchers discovered that significant subjective and physiological changes were not observed until a minimum of 3 h of exposure to standing, and suggested longer testing durations. Findings from a review by Halim and Omar (2011) and a review by Waters and Dick (2015) also supported the use of anti-fatigue mats and ergonomic footwear. Mohd Noor et al. (2013) found limited evidence that supports the effectiveness of these interventions in real industrial workplaces. Age and gender effects were not addressed in the literature to date.

Standing related occupational injuries come at a cost to both the individuals and the organisations, attributed to decrements in performance and productivity, absenteeism, medical costs and worker health and well-being (Ahmad et al., 2006). It is therefore important to seek clarification about the benefits of currently utilised interventions in order to target risk control strategies that maximise worker health and well-being and consequently profitability to organisations.

This review aims to deliver a methodologically and rigorously sound evaluation of literature on the effectiveness of softer flooring, anti-fatigue mats and shoe insoles on a range of outcomes related to discomfort and fatigue in those who stand for prolonged periods at work. It will also consider longer exposure periods as recommended by Redfern and Cham (2000). The review will also attempt to assess the impact of age and gender on the outcomes of prolonged standing as these important factors remain largely unexplored, yet there is evidence that a greater proportion of men than women usually stand at work (Tissot et al., 2009) but more women report work-related musculoskeletal pain (Messing et al., 2015). The ageing workforce adds further complexity to the issue of prolonged standing at work. In Australia for example, labour force data indicates a sharp increase in the representation of workers who are 45 years and older in the public sector since 1998 (Public Sector Commission, 2010). Research exploring age related injury factors confirms that older workers are more susceptible to injury and suffer delayed recovery (Quinlan, 2010).

This review will attempt to fill the existing gap in research and help to substantiate and further inform industry recommendations and future research directions in this area, by answering the following research questions:

- Does a cushioning material, compared to a hard floor, reduce lower body musculoskeletal discomfort in workers who work in a constrained standing position for prolonged periods?
- Are cushioned surfaces more effective than shoe inserts in reducing musculoskeletal discomfort of the lower body?
- Is there an age difference in the effectiveness of cushioning materials on musculoskeletal discomfort?
- Is there a gender difference in the effectiveness of cushioning materials on musculoskeletal discomfort?

2. Methods

The review was registered with the International Prospective Register of Systematic Reviews (PROSPERO), with registration number CRD42016039442. The registration can be accessed at http://www.crd.york.ac.uk/PROSPERO.

2.1. Search strategy

Problem/Population, Intervention, Comparison and Outcome (PICO) plus Setting model was selected to frame the research questions and the methodology (Huang et al., 2006; Methley et al., 2014).

To identify the relevant peer reviewed studies, a search of seven electronic databases was performed using a pre-determined set of search keywords, developed in collaboration with a content specialist librarian (Table 1).

The relevant databases included Medline, EMBASE, CINAHL, PsychInfo, Web of Science, PubMed and Cochrane Library. In addition, in order to minimise the risk of publication bias, an internet search of grey literature including conference proceedings, was performed using Google Scholar. A search of bibliographies of the included articles was conducted and cited references for included articles were searched using Web of Science. Advice from two subject matter experts was sought via email communication with regards to the completeness of the search results.

The screening for eligibility process occurred in two stages. The relevant studies were screened by title and abstract in the first instance, then by full text against the inclusion and exclusion criteria. Two researchers (GS and TK) were independently involved in the screening process and resolved any differences through discussion. A third author (KH) was available to arbitrate if consensus regarding eligibility was not reached.

Once all relevant studies were identified, a data extraction form was designed.

Table 1
List of search keywords.

Concept	Keywords
1.Standing posture at work (population)	Standing OR "upright position" OR "stand* to work" OR "work posture*" OR "stand upright" OR "stand up"
2.Standing Surface (intervention & comparative intervention)	"soft floor*" OR "soft surface*" OR cushion* OR "cushioning insole*" OR "anti fatigue mat*" OR "floor mat*" OR "shoe orthotic*" OR "foot orthosis" OR "foot orthoses" OR shoe* OR "hard floor*" OR "hard surface*" OR "solid floor*" OR "concrete floor*" OR "concrete surface*"
3.Musculoskeletal response (outcome)	"muscle fatigue" OR "muscle discomfort" OR "back pain" OR "spinal pain" OR "leg pain" OR "musculoskeletal symptom*" OR "musculoskeletal pain" OR "musculoskeletal disorder*" OR "musculoskeletal discomfort*" OR "musculoskeletal comfort" OR "work related musculoskeletal disorder*" OR "work related musculoskeletal discomfort*" OR WRMSD OR WMSD OR MSD OR "musculoskeletal syndrome" OR "occupational disease*" OR "low* back pain*" OR LBP OR "work related fatigue" OR sciatica OR "musculoskeletal disease* risk factor*" OR "foot volume" OR "foot swelling" OR "leg volume" OR "leg swelling" OR "leg circumference" OR "foot circumference" OR "force plate" OR "leg o?dema*" OR "foot o?dema" OR "weight shift*" OR "centre of pressure" OR "muscle twitch force" OR electromyography OR EMG OR "skin temperature" OR "biomechanical response*" OR "biomechanical measure*" OR "physiological measure*" OR "physiological response*" OR "muscle co activation" OR "muscle coactivation"

Using guidance provided by the Cochrane Collaboration (Higgins and Green, 2011). The data extraction form was used to document study characteristics and findings.

2.2. Eligibility criteria

2.2.1. Inclusion criteria

All five elements of the PICO + Setting model were considered in the selection of inclusion and exclusion criteria. The review included peer reviewed empirical studies of healthy participants of working age (18-65 years) who performed tasks in constrained standing posture for a minimum of 3 h (Population), with subjects using a cushioning interface between the floor and the foot (Intervention) versus the subjects standing on a hard floor (Comparison). Studies conducted in real workplaces and those conducted in simulated work environments were included (Setting). Included studies explored at least one of the following outcomes: subjective discomfort/fatigue ratings, objective local muscle fatigue, Centre of Pressure (COP)/weight shifts and leg volume (Outcome).

The search was limited to studies in the English language. The most relevant previous review on this topic by Redfern and Cham (2000) reported on the results of studies prior to year 2000. This review builds on those findings by examining literature published between January 2000 and April 2016.

2.2.2. Exclusion criteria

Studies of children, adolescents and the elderly as well as those, which specifically selected study populations with pre-existing musculoskeletal, neurological or other conditions were excluded from the review. Interventions of lesser duration than 3 h and those, which deviated from constrained standing postures such as interventions, which considered walking or running, were also excluded. Interventions that analysed the effects of standing surfaces other than those characterised by cushioning properties, for example sloped surfaces or magnetic inserts, instability shoes or shoes with high heels were excluded.

2.3. Risk of bias assessment

Two researchers (GS and TK) independently performed the risk of bias assessment for the included studies. A third researcher was available to adjudicate when consensus could not be reached (KH). The majority of the studies included in the review were cross-over trials, where the participants are subjected to a number of interventions and act as their own controls. Appropriate time period between interventions is required in order to minimise the carry over effect from one intervention to the next. This type of study design is suitable for evaluating interventions that cause a temporary effect in the treatment of stable or chronic conditions (Ding et al., 2015). A risk of bias tool designed specifically to assess cross-over studies was used as recommended by Ding et al. (2015). The specific risk of bias domains suggested by Ding et al. (2015) included appropriateness of cross-over design, randomisation of treatment order, carry-over effect, data collection bias, allocation concealment, blinding, incomplete outcome data, selective outcome reporting and other bias. Each of these nine domains was given a rating of low, unclear or high based on specific definitions. The risk of bias of the two remaining pre-post intervention trials were assessed using the Cochrane risk of bias tool developed by the Cochrane Collaboration (Higgins and Green, 2011).

2.4. Quality of evidence assessment

The levels of evidence rating scale tool as proposed by van Tulder et al. (2003) was selected to assess the quality of evidence for each identified outcome. This is an assessment of evidence tool, which is based on the design of the included studies. More complex evidence rating tools such as the Grading of Recommendations, Assessment, Development and Evaluation (GRADE) approach (Guyatt et al., 2008) were judged not to be appropriate because of the impact specific study design has on the level of quality of evidence. The van Tulder scale considers alternative study designs including those included in this review. The scale rates the level of evidence from strong, through to moderate, limited, conflicting and no evidence as per the definitions in Fig. 1.

Level of Evidence	Definition
Strong	Consistent findings among multiple high quality RCTs*
Moderate	Consistent findings among multiple low quality RCTs and/or CCTs** and/or one high quality RCT
Limited	One low quality RCT and/or CCT
Conflicting	Inconsistent findings among multiple trials (RCTs and/or CCTs)
No evidence	No RCTs or CCTs

*RCT = Randomised Controlled Trial.
**CCT = Controlled Clinical Trial.

3. Results

Data from included studies were synthesised based on the outcomes being investigated. Risk of bias and quality of evidence were examined for each outcome in order to answer the research question.

The electronic database search initially generated 946 manuscripts. Following the removal of 71 duplicates, 875 manuscripts were screened based on their title and abstract, of which 46 manuscripts were selected for full text review. One additional manuscript was retrieved from the

Level of Evidence	Definition
Strong	Consistent findings among multiple high quality RCTs*
Moderate	Consistent findings among multiple low quality RCTs and/or CCTs** and/or one high quality RCT
Limited	One low quality RCT and/or CCT
Conflicting	Inconsistent findings among multiple trials (RCTs and/or CCTs)
No evidence	No RCTs or CCTs

* RCT = Randomised Controlled Trial

** CCT = Controlled Clinical Trial

Fig. 1. Levels of evidence. Taken from 'Updated method guidelines for systematic reviews in the Cochrane Collaboration Back Review Group' by M. van Tulder et al., 2003, *Spine*, 28 (12), p. 1296.

bibliographic and cited reference search and four from other sources (internet search). Following full text review, a further 39 manuscripts, including three manuscripts from one cross-sectional study, were excluded because they did not meet the inclusion criteria or were not in an appropriate study design. The quality of data collected from cross-sectional studies can be limited, because the direction of effect is uncertain and the outcome reporting reliant on recall rather than direct observation or measurement (Bonita et al., 2006). Due to the limitations of a cross-sectional study design, we decided not to include these manuscripts in the review.

Three manuscripts were related to larger doctorate research on the effect of flooring surface compliance on plantar pressures and discomfort during prolonged standing (Wiggermann, 2011; Wiggermann and Keyserling, 2010, 2013). Data from the Wiggermann and Keyserling (2013) manuscript were included, due to the specific relevance of this research experiment to the review. Two manuscripts examined different outcomes from a single study (Orlando and King, 2004; Zander et al., 2004) and one manuscript described two different experiments (Lin, Chen and Cho, 2012b). In summary, the systematic search located 10 unique studies from 12 separate manuscripts for analysis in this review (Fig. 2).

3.1. Participants

A total of 361 participants were involved across all 10 studies from the 12 manuscripts. Six were from real workplaces and four from simulated workplaces. The simulated workplace experiments had relatively smaller sample sizes of 10–16 compared to real workplace experiments, which had as many as 153 participants in the Wahlstrom et al. (2012) study. Female participants dominated the overall gender demographic in the selected studies with 233 female participants compared to 128 male participants.

Age as a demographic was not reported consistently. Specific age related data were missing in four studies (Garcia-Hernandez et al., 2016; Jefferson, 2013; Wahlstrom et al., 2012; Zander et al., 2004). From the remaining six studies, overall younger workers ($\leq$45 years) contributed 56% and older workers ($\geq$45 years) contributed 44% to the review results.

3.2. Study characteristics

The information from the data extraction form included study title and author, study design, study setting, the number and the characteristics of study participants including age and gender, the nature of interventions tested, measurement tools used and outcomes. The information about study characteristics and findings is incorporated within the outcome tables (Tables 3–5 and Table 6).

3.3. Types of interventions

Many of the studies examined different interventions making comparisons of results between studies challenging. For example, Cham and Redfern (2001) and Wiggermann and Keyserling (2013) studied a selection of mats made of various material properties. Brownie and Martin (2015) compared the effects of a rubber mat, sole insert and linoleum tiles on concrete. The Orlando and King (2004) and Zander et al. (2004) field study compared the effects of a mat against the effects of insoles; whilst King (2002) explored the effects of a combined condition of standing on a mat wearing insoles. Garcia-Hernandez et al. (2016) compared usual footwear with the effects of wearing customised orthoses whilst Jefferson (2013) compared usual footwear with Sorbothane shoe insoles on concrete floor. In the laboratory experiment, Lin et al. (2012b) took a different approach and compared working barefoot and wearing sports shoes on a hard floor and on a mat. The study by Wahlstrom et al. (2012) was also unique in that two different floor covering conditions were evaluated in two different aged care centres.

Not only did the studies apply different interventions, the control conditions also varied between studies and included vinyl covered steel, concrete, wood, and linoleum tile. In one study the type of hard floor surface was not stated (Garcia-Hernandez et al., 2016). The studies also reported on a range of different outcomes. The variations in the intervention floor types, control conditions, as well as the range of different outcomes, make direct comparisons of the effects implausible. As a result, it was not feasible to conduct a meta-analysis of these studies or a subgroup of the results.

3.4. Risk of bias

Overall, the risk of bias assessment revealed suitable study design and complete data collection and reporting in the selected studies. Most of the studies applied interventions in a randomised order. The risk of a carry-over effect was probably insignificant, because the interventions only had a short lasting effect and were conducted on non-consecutive days however, none of the studies actually evaluated the carry over effect as a potential source of bias as recommended by Ding et al. (2015). Carry-over effect occurs in circumstances when, due to an inadequate period between interventions, the previous intervention interferes with the outcome of the subsequent intervention (Higgins and Green, 2011). Another potential source of bias among the studies was associated with the lack of participant blinding. In experiments that compared cushioned flooring or mats with shoe insoles, participant blinding is difficult. However, in the case of various floor surface comparisons, at least one study succeeded in blinding by effectively covering up the standing surface (Aghazadeh et al., 2015). Additional potential sources of bias identified amongst the included studies were: variation amongst the usual participant standing habits (Cham and

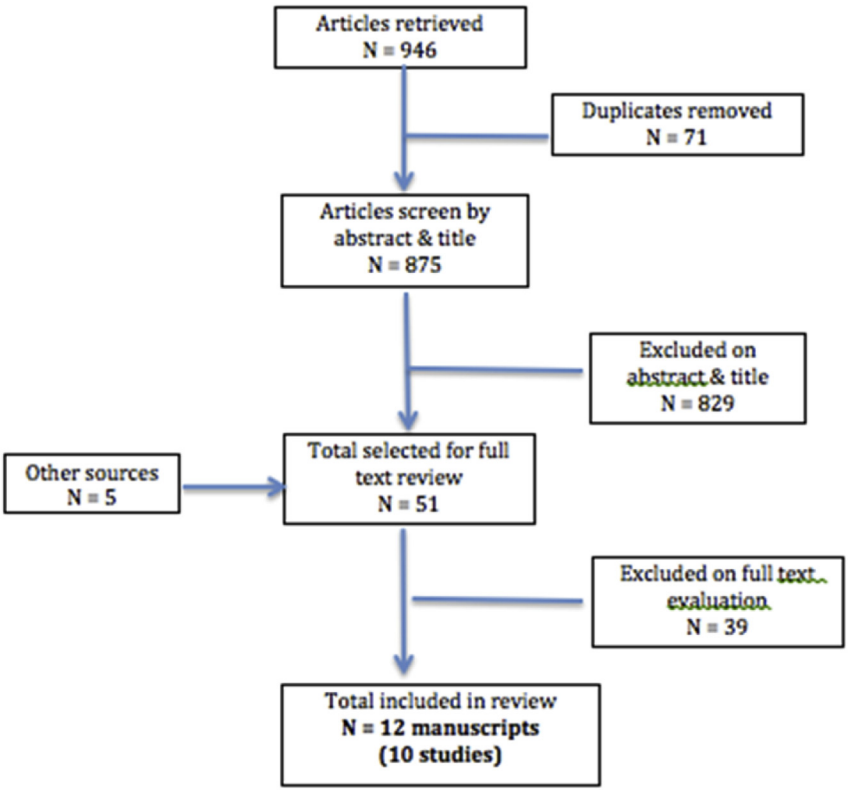


Fig. 2. Flowchart of search results and inclusion.

Redfern, 2001; Garcia-Hernandez et al., 2016; Jefferson, 2013; King, 2002; Orlando and King, 2004); non-standardised footwear (King, 2002; Lin et al., 2012b; Orlando and King, 2004; Zander et al., 2004); and undefined levels of physical activity or oral food and drink intake in the days prior to testing. Age and gender were also likely sources of bias in some studies. For example, Brownie and Martin (2015) included more younger men and older women in their sample and attempted comparisons between the younger and the older group. Garcia-Hernandez et al. (2016), Jefferson (2013) and the laboratory experiment by Lin et al. (2012b) included a much higher ratio of male participants than female participants and the opposite was true in the studies by Orlando and King (2004) and Zander et al. (2004) whilst only female participants were involved in Wahlstrom et al. (2012) (Table 2, Fig. 3 and Table 3).

3.5. Outcome measures

In order to obtain meaningful analysis, the data were organised and examined by outcome. Four specific analyses were conducted. These included: the effect of cushioning surfaces on subjective discomfort/fatigue, local muscle fatigue, Centre of Pressure (COP)/weight shifts and leg volume. A comparison of the effectiveness of mats versus insoles as well as data related to age and gender effects were also reviewed and addressed.

3.5.1. Subjective discomfort/fatigue

Nine studies examined the effect of standing surfaces on musculoskeletal discomfort and fatigue (Table 4). Four were in a simulated workplace environment and five in a real workplace environment. Various discomfort-rating scales were used (Table 4). Consistently the studies found that discomfort generally increased with standing time. Various treatment conditions appeared to reduce discomfort compared to standing on hard surfaces. However, of the six studies, which compared the effect of a number of interventions (Brownie and Martin, 2015; Cham and Redfern, 2001; King, 2002; Lin, Chen and Cho, 2012a; Orlando and King, 2004; Wiggermann and Keyserling, 2013), only three demonstrated a statistically significant impact of one intervention versus another in the reduction of discomfort (Brownie and Martin, 2015; Cham and Redfern, 2001; Lin et al., 2012a). King (2002), Orlando and King (2004) and Wiggermann and Keyserling (2013) found no statistically significant differences between flooring conditions. As there are consistent findings from CCTs according to the van Tulder et al. (2003) criteria, there is moderate level of evidence for a reduction of discomfort and/or fatigue when using a cushioned surface for prolonged standing work. However, conflicting evidence of an effect of specific materials exists as demonstrated by three CCTs delivering statistically significant results and three CCTs providing no statistically significant impact (van Tulder et al., 2003).

3.5.2. Local muscle fatigue

Local muscle fatigue, that is a reduction in muscle twitch force (MTF), was measured using EMG in two simulated workplace studies. Although there were substantial differences between the design methodologies, the two studies investigated the effects of various floor surfaces on local muscle fatigue (Brownie and Martin, 2015; Cham and Redfern, 2001). Neither study found an association (Table 5). A decrease in muscle twitch force was found by Brownie and Martin (2015) in all testing conditions in both the younger and older participants, suggesting that local muscle fatigue is associated with prolonged standing regardless of the standing surface. As there were consistent findings between the two CCTs, based on the criteria by van Tulder et al. (2003), there is moderate level of evidence that prolonged standing on a softer surface or using cushioned insoles has no effect on lower limb muscle fatigue.

3.5.3. COP/weight shifts

Mixed results were found between the three simulated workplace studies that investigated COP/weight shift (Table 6). The studies used

Table 2
Risk of bias assessment of crossover studies.

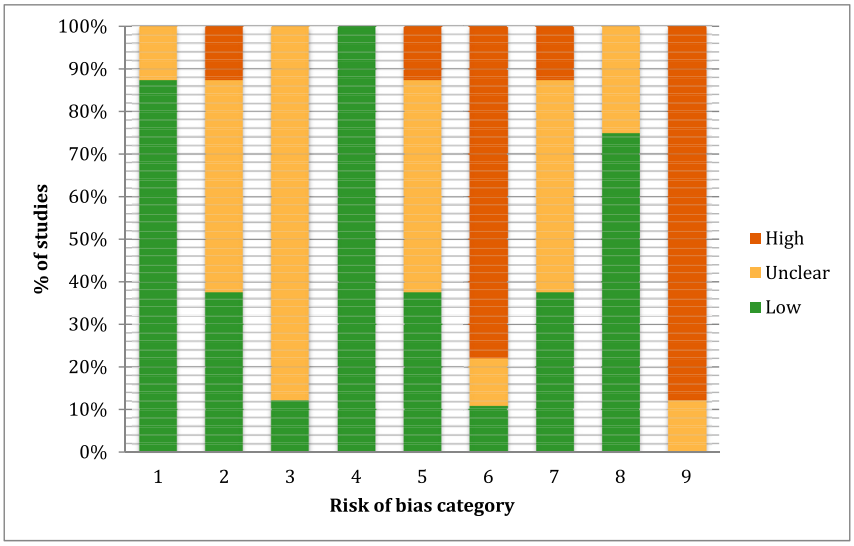
Risk Of Bias	Brownie and Martin 2015	Cham and Redfern 2001	Garcia-Hernandes et al., 2016	Jefferson 2013	King 2002	*Lin et al., 2012a,b Exp 1	**Lin et al., 2012a,b Exp 2	***Orlando and King 2004 ****Zander et al., 2004	Wahlstrom et al., 2012	Wiggermann and Keyserling 2013
1. Appropriate cross-over design	Low	Low	NOT Cross-over study design See table 3	Unclear	Low	Low	Low	Low	NOT Cross-over study design See table 3	Low
2. Randomised treatment order	Unclear	Low		High	Low	Unclear	Unclear	Low		Unclear
3. Carry-over effect	Unclear	Unclear		Low	Unclear	Unclear	Unclear	Unclear		Unclear
4. Unbiased data	Low	Low		Low	Low	Low	Low	Low		Low
5. Allocation concealment	Unclear	Unclear		High	Low	Unclear	Unclear	Low		Low
6. Blinding	High	High		High	High	High	High	***High ****Low		
7. Incomplete outcome data	Unclear	Low		Low	Unclear	High	Low	Unclear		Unclear
8. Selective outcome reporting	Low	Low		Unclear	Low	Unclear	Low	Low		Low
9. Other bias	Unclear	High		High	High	High	High	High		High

* Lin et al., 2012a,b Exp 1 describes Experiment 1.
** Lin et al., 2012a,b Exp 2 describes Experiment 2.
*** Orlando and King, 2004 & ****Zander et al., 2004 describe results from the same study; discrepancy in the blinding of participants and personnel risk of bias exists and concerns the measurement of different outcomes.

Table 3
Risk of bias assessment of non-crossover studies.

Risk Of Bias	Brownie and Martin 2015	Cham and Redfern 2001	Garcia-Hernandes et al., 2016	Jefferson 2013	King 2002	*Lin et al., 2012a,b Exp 1	**Lin et al., 2012a,b Exp 2	***Orlando and King 2004 ****Zander et al., 2004	Wahlstrom et al., 2012	Wiggermann and Keyserling 2013
Random sequence generation	Cross-over study design See Table 2	Cross-over study design See Table 2	N/A	Cross-over study design See Table 2	Cross-over study design See Table 2	Cross-over study design See Table 2	Cross-over study design See Table 2	Cross-over study design See Table 2	High	Cross-over study design See Table 2
Allocation concealment			N/A						High	
Blinding of participants and personnel			High						High	
Blinding of outcome assessment			N/A						High	
Incomplete outcome data			Low						High	
Selective reporting			Low						Low	
Other bias			Unclear						High	

* Lin et al., 2012a,b Exp 1 describes Experiment 1.
** Lin et al., 2012a,b Exp 2 describes Experiment 2.
*** Orlando and King, 2004 & ****Zander et al., 2004 describe results from the same study; discrepancy in the blinding of participants and personnel risk of bias exists and concerns the measurement of different outcomes.



Note: numbers refer to the category of risk of bias as per table 2

Fig. 3. Graphical representation of risk of bias results for crossover studies.

different measurement methods, making comparisons challenging. Cham and Redfern (2001) and Lin et al. (2012a) used the force platform to assess COP, whereas Wiggermann and Keyserling (2013) used in-shoe pressure sensors. In their laboratory experiment, Lin et al. (2012a) did not find a link between the number of weight shifts and floor or footwear conditions despite using a more sensitive measurement than Cham and Redfern (2001). Cham and Redfern (2001) found a statistically significant increase in the number of weight shifts only in the 4th hour of testing with the softest and the hardest mats. They found a correlation between the material properties of mats and weight shifts and concluded that increased elasticity, decreased energy absorption and increased stiffness properties of mats have a bearing on decreasing weight shifts. Similarly, Wiggermann and Keyserling (2013) found a statistical difference between the mats and weight shifts. In their study the hardest mat produced most weight shifts and the greatest COP excursion. The authors found a positive correlation between weight shifts and discomfort but in contrast to Cham and Redfern (2001), they found no relationship between the material properties of stiffness and work lost and weight shifts. Consequently, according to the van Tulder et al. (2003) criteria, there is conflicting evidence for the relationship between weight shifts and softer flooring.

3.5.4. Leg volume

Four studies (two simulated workplace and two real workplace) measured changes in lower leg volume. Cham and Redfern (2001) used the water tank displacement method whilst Zander et al. (2004) and Lin et al. (2012a) in both experiments used the Gulick tape to measure the changes in shank and thigh circumference. All four studies showed that leg swelling increases with standing time regardless of the flooring condition (Table 7). However, not all were able to demonstrate the benefit of cushioned surfaces on reducing the magnitude of the effect. The testing periods varied from 4 h in the studies by Cham and Redfern (2001), and Lin et al. (2012a) to 8 h in the study by Zander et al. (2004). The pre and post testing measures taken do not provide insight into the effects after 4 h in the study by Zander et al. (2004) to make effective comparisons between the studies. For example, the results of both experiments by Lin et al. (2012a) supported the use of anti-fatigue mats in limiting the increase in leg volume from prolonged standing. In contrast, Cham and Redfern (2001) and Zander et al. (2004) found no statistically significant difference between the various intervention conditions. As there are inconsistent findings amongst the CCTs, there is conflicting evidence for the effect of cushioned surfaces on leg swelling based on the criteria by van Tulder et al. (2003).

3.6. Mats versus insoles comparisons

Four studies (two studies in simulated workplaces and two studies in real workplaces) attempted to compare the effects of standing on a mat and standing wearing cushioned insoles (Brownie and Martin, 2015; King, 2002; Lin et al., 2012b; Orlando and King, 2004) (Table 4). There was a trend towards insoles producing slightly greater benefits compared to mats (Brownie and Martin, 2015; King, 2002; Orlando and King, 2004), although only Brownie and Martin (2015) reached statistical significance. As only one CCT delivered a statistically significant outcome according to the van Tulder et al. (2003) criteria, there is only limited level of evidence in favour of insoles.

King (2002) and Lin et al. (2012b) additionally investigated the effect of combined conditions of using insoles/sport shoes whilst standing on mats and discovered that it had a greater effect on reducing discomfort than wearing inserts or standing on a mat alone. Consistent findings amongst these two CCTs provide a moderate level of evidence as per the criteria by van Tulder et al. (2003) to support the use of both simultaneously.

Another discovery of interest is a faster recovery from local muscle fatigue 1 h post standing demonstrated by Brownie and Martin (2015) when wearing insoles compared to standing on a mat. This finding is based on a result from one relatively small CCT providing a limited level of evidence based on the van Tulder et al. (2003) criteria and should be confirmed by further research in the field conducted over a longer period.

3.7. Age and gender effects

Limited age and gender related data were presented in the included studies, providing little information on which to base meaningful conclusions. Specific age related data were missing in one study (Jefferson, 2013) and only mean age was provided in the studies by Garcia-Hernandez et al. (2016) and Wiggermann and Keyserling (2013). Of those studies that identified the age of the participants, Cham and Redfern (2001), Lin et al. (2012b) and Wiggermann and Keyserling (2013) studied younger groups and King (2002) and Orlando and King (2004) studied older groups. Those studies, which investigated the effects of age (Brownie and Martin, 2015) or gender (King, 2002) on discomfort/fatigue, found no significant association. The analysis of

Table 4
Pooled analysis: subjective discomfort & fatigue.

Outcome: Subjective discomfort & fatigue									
	Brownie and Martin, 2015	Cham and Redfern, 2001	Garcia-Hernandez et al., 2016	Jefferson, 2013	King, 2002	Lin et al., 2012a,b (Experiment 1)	Orlando and King, 2004	Wahlstrom et al., 2012	Wiggermann and Keyserling, 2010
Study design	Cross-over trial	Cross-over trial	Pre-post design	Cross-over trial	Cross-over trial	Cross-over trial	Cross-over trial	Pre-post design with control group	Cross-over trial
Setting	Laboratory	Laboratory	Field	Field	Field	Laboratory	Field	Field	Laboratory
Age	Young (n = 10)	Young	Mean age = 42.9 yrs	Not reported	Old	Young	Old	(I): 20-64 yrs (mean 46 yrs)	Young
Young (< 45yrs) *Old (> 45yrs)*	Old (n = 6)							(C): 24-65 yrs (mean 47 yrs)	
Gender	M = 9 F = 7	M = 5 F = 5	M = 39 F = 9	M = 38 F = 2	M = 5 F = 17	M = 10 F = 3	M = 2 F = 9	M = 0 F = 153	M = 5 F = 5
Design	5 h standing	4 h standing	8 wks; pre-post	4 × 12 h shifts	1 wk × 8 h shifts	4 h standing	8 h shift	f/u over 24 months	4 h standing
Breaks	5min end of 1, 2, 4 h + 30min end of 3rd hr	2min walk every 30min	Unspecified	Unspecified	Unspecified	10 min at end of each hr sitting	Unspecified	Unspecified	10min walk/sit (half way through testing)
Interventions	Rubber mat ¾ inch thick (19.05 mm), hardness 50-55 Insole	6 mats (thickness): B = 7.1 mm C = 7.1 mm D = 11.2 mm E = 15.6 mm F = 16.9 mm G = 14.6 mm	Customised orthoses (Podoactiva)	Insoles (Sorbothane) in work shoes without arch support	Mat 5/8 inch thick (15.9 mm) Insoles (Viscolas) Mat + Insoles	Hard floor (force plate) + barefoot Hard floor + sport shoes Mat (12.5 mm thick) + barefoot Mat + sport shoes	Mat 5/8 inch thick (15.9 mm) Insoles (Viscolas)	4 mm vinyl floor (1.5 mm thick with 2.5 mm foam backing)	4 mats A (softest) B (softer) C (harder) D (hardest)
Footwear	Standardised sports shoes	Standardised hard sole shoes	Unspecified	Normal footwear	Not standardised steel cap shoes	Sport shoes	Not standardised work shoes	Not standardised	Standardised sports shoes (New Balance)
Control surface	Linoleum tiles on concrete	Vinyl tile on steel plate (floor A)	Unspecified	Concrete floor	Wood floor	Force plate	Wood floor	2.5 mm linoleum floor	Linoleum tile on concrete
Testing times/ frequency	Pre, half way through and post	Pre & at end of each hr	Pre (T1), at 4 wks (T2) & post (T3) at 8 wks	At end of each shift	At end of Mon, Wed & Fri	Pre & at end of each hr	End of shift	4 wks prior to intervention; 6 wks, 12 & 24 months post intervention	Pre & at end of each hr
Testing method	VAS: Sites: - feet - lower legs - knees - upper legs - buttocks - Low back	CR10 Borg Scale: Discomfort sites: - feet - ankles - lower legs - knees - upper legs - hips - low back - upper back Perceived firmness of floor General fatigue Entire leg fatigue		Body discomfort chart (0-10): Sites: - feet - knees - low back/ buttocks - neck/ Upper back - shoulders/upper arms - hands	5-point Likert scale: Discomfort sites: - feet - ankles - lower legs - knees - upper legs - hips - low back - upper back Perceived firmness of standing surface General fatigue Leg fatigue Preference to continue using the intervention (Y/N)	CR10 Borg Scale (7-point Likert Scale): Leg discomfort	5-point Likert scale: Discomfort sites: - feet - ankles - lower legs - knees - upper legs - hips - low back - upper back Perceived firmness of standing surface General fatigue Leg fatigue Preference to continue using the intervention (Y/N)		VAS: Sites: - feet - lower legs - knees - thighs - buttocks - low back

(*continued on next page*)

Table 4 (*continued*)

Outcome: Subjective discomfort & fatigue									
	Brownie and Martin, 2015	Cham and Redfern, 2001	Garcia-Hernandez et al., 2016	Jefferson, 2013	King, 2002	Lin et al., 2012a,b (Experiment 1)	Orlando and King, 2004	Wahlstrom et al., 2012	Wiggermann and Keyserling, 2010
			NMQ Sites: - ankles/feet - knees - hips/thighs - low back - upper back - neck - shoulders - elbows - wrists/hands Additional questions in T2 & T3 (5-point Likert scale T1: - MSDs in past 12 months T2: - comfort with insoles - general fatigue - leg fatigue - preference T3: - adaptation period length of time - perception of benefit from insoles					Borg CR 10 Scale: Local fatigue in 4 body sites: - feet - knees - hips - low back Pain intensity (11-point scale): - current - worst in last month - average in last month in 4 body sites (mean of 3 ratings for each body site) Pain related disability in 4 body sites for (11-point scale): - daily activities - social & family activities - work (mean of 3 ratings for each body site)	
Results	Statistically significant reduction in foot discomfort with insoles than mats or linoleum No significant age differences on discomfort or fatigue ratings	Statistically significant difference in discomfort and leg fatigue after 3rd hour for all but one body parts (not upper back) ($p < 0.05$) Correlation was found between reduced discomfort/fatigue and mats with increased elasticity, decreased energy absorption and increased stiffness ($r = 0.4–0.9$)	Statistically significant reduction in MSD issues in 7/9 body sites (all but elbows and wrists/hands) Perceived improvement: 0 – worse 7 (14.6%) - no improvement 41 (85.4%) - improvement <u>Preference:</u> At end of 8 wks 98% participants preferred to use orthoses			Statistically significant influence of floor condition and time standing on discomfort ($p < 0.01$) Mat + sport shoes: least uncomfortable condition Hard floor + barefoot: most uncomfortable condition Greater influence of floor type than shoe condition		Statistically significant reduction in pain intensity and pain-related disability and perceived exertion in feet even after 2 yrs ($p = 0.05$)and pain intensity after 2 years in low back ($p = 0.05$) *(note: response rate at 2 yrs = 73%; n = 112)*	

(*continued on next page*)

Table 4 (*continued*)

Outcome: Subjective discomfort & fatigue									
	Brownie and Martin, 2015	Cham and Redfern, 2001	Garcia-Hernandez et al., 2016	Jefferson, 2013	King, 2002	Lin et al., 2012a,b (Experiment 1)	Orlando and King, 2004	Wahlstrom et al., 2012	Wiggermann and Keyserling, 2010
				Statistically significant decrease in discomfort scores ($p = 0.0001$) Mean low back pain score 28% with insoles (2.86\2.06) Mean foot pain score 32% with insoles (5.09\3.48) Reduction of knee pain by 38% (data not presented) No correlation between change in low back pain, knee pain and the change in foot pain with insoles ($r = 0.148$) Slight increase in low back pain and foot pain each day with normal footwear. Lower discomfort level and no increase each day with insoles	Statistically significant difference in discomfort between wood floor & insoles; wood floor & combined (mat + insoles) condition No statistically significant difference between intervention conditions Data indicates overall lowest discomfort scores with combined condition followed by insoles then mats For knees, hips and low back – no statistically significant difference in discomfort ratings with mat but statistically significant with insoles and combined condition No significant gender differences on discomfort or fatigue ratings Preference: Least for wood floor		No statistically significant difference between flooring conditions Lower mean scores with mat and insoles compared to wood floor Lower mean scores for general and leg fatigue with insoles but did not reach statistical significance Insoles – most comfortable in feet, ankles and low back Mat – most comfortable in lower legs, knees, upper legs and upper back but did not reach statistical significance Perception of firmness: Statistically significant between mat &/or insole versus wood, but NOT between mat and insoles Preference: Greatest for insoles (91%); than mat (67%) and wood floor (20%)		Discomfort reduced by 3 mats compared to control but no significant difference among mats Significant flooring effects on body sites other than the lower leg only in 4th hr Statistically significant effect of flooring in 1st, 2nd & 3rd hr only on lower leg Significantly lower discomfort ($p < 0.05$) for 'overall leg' with A & B Significantly lower discomfort for feet with A No statistical significance ($p = 0.07$) for low back or knees No effect of flooring on overall body, buttocks or thighs Only in feet increasing stiffness and decreasing work lost properties of mats correlated with increased discomfort

Table 5
Pooled analysis: local muscle fatigue.

Outcome: Local muscle fatigue		
	Brownie and Martin, 2015	Cham and Redfern, 2001
Study design	Cross-over trial	Cross-over trial
Setting	Laboratory	Laboratory
Age *Young = < 45 yrs;* *Old = > 45 yrs*	Young (n = 10) Old (n = 6)	Young
Gender	M = 9 F = 7	M = 5 F = 5
Design	5 h standing	4 h standing
Breaks	5min end of 1, 2, 4 h + 30min end of 3rd hr	2min walk every 30min
Interventions	Rubber mat ¾ inch thick (19.05 mm), hardness 50-55 Insole	6 mats (thickness): B = 7.1 mm C = 7.1 mm D = 11.2 mm E = 15.6 mm F = 16.9 mm G = 14.6 mm
Footwear	Standardised sport shoes	Standardised hard sole shoes
Control surface	Linoleum tiles on concrete	Vinyl tile on steel plate (floor A)
Testing times	Before test, at mid-point (ie 2.5 h from commencement), straight after, 30min after, 60min after	Measured every 15 - 20min
Testing method	Sitting L leg on foot plate Submax stimulation of gastrocs/soleus Stimulation intensity = 30-40 mA, pulse duration 1ms, freq 2 Hz	Over 1 min at sampling rate of 1 kHz Electrodes over tibialis anterior & soleus R leg & R & L erector spinae
Results	No effect of mat (or insole) on preventing or reducing the development of fatigue A significant decrease in MTF ($p < 0.05$) for younger & older participants (between age groups not significant ($p > 0.05$) with prolonged standing Recovery quicker in younger participants Younger: recovery after 1 h better with insoles than mat Older: similar recovery with all surfaces	No statistically significant change in median frequency of the EMG power spectrum

data by Brownie and Martin (2015) showed that younger workers recover quicker than older workers. As there is only evidence from one study on the effects of age and one study on the effects of gender on the outcomes from using cushioned standing surfaces, there is limited level of evidence of no effect according to the van Tulder et al. (2003) criteria.

4. Discussion

The aim of this review was to gather and appraise all relevant literature on the effects of using anti-fatigue matting, soft flooring or shoe insoles on musculoskeletal discomfort whilst standing at work. The effects of these interventions on a number of discomfort related outcomes were analysed, including subjective reporting of discomfort/fatigue, physiological changes in local muscle fatigue and leg volume and behavioural responses characterised by weight shifting. Our overall findings are synthesised and presented below.

The search identified a limited number of studies, which sufficiently met the inclusion criteria. The included studies were heterogeneous in terms of testing protocols, interventions, control conditions and methods of measuring outcomes, making it infeasible to conduct a meaningful meta-analysis. In addition, there were no randomised controlled trials (RCTs). This combined with the highly variable risk of bias results amongst the included studies decreases the level of confidence with which determination of effect can be made. However, some consistent trends have been identified to help inform industry recommendations and shape future research directions in this area.

Overall, cushioned materials are likely to reduce discomfort and reported fatigue for workers who are required to stand for substantial periods of time at work. However, very limited evidence was found in support of specific material properties as important factors when choosing the optimum type of surface. Until further research delivers stronger evidence for the benefit of using specific cushioning materials, other factors such as cost, durability, convenience and safety may be more appropriate considerations than material properties for workplaces when deciding on the type of cushioned flooring.

Whilst some evidence exists for the reduction of discomfort from using softer flooring, there is much less certainty about the effect softer flooring has on physiological outcomes. The reasons are unclear, but several theories have been proposed. One explanation relates to the limited sensitivity of some of the currently available tools in measuring the subtle changes in outcomes. For example, Cham and Redfern (2001) suggested that EMG is not a sensitive tool for the assessment of the effect of flooring conditions on local muscle fatigue in prolonged standing, because muscle forces produced during standing are lower compared to forces generated during dynamic tasks, making them more difficult to detect via EMG.

Variability in study designs and limitations in controlling for confounding variables might offer another explanation for the inconsistencies in results. Commonly, footwear and oral intake were not standardised in the experiments. Fluid and salt intake together with well functioning internal body systems, play an important role in maintaining a healthy fluid balance in the body (Sherwood, 1997). Failure to measure and monitor the body's haemodynamic environment and oral intake may lead to inaccurate interpretation of leg volume results. With this in mind, the relative contributions to changes in leg volume from fluctuations in the internal body fluid balance compared to changes from prolonged standing on various surfaces cannot be established with confidence, and should be a factor in future studies. Similarly, factors in these studies may exert influence on one another, for example, discomfort may lead to increased number of weight shifts and increased leg volume may contribute to discomfort. However, judgments about the specific relationships between them and the effect of one on another are difficult to establish from such a small pool of studies.

These inconsistencies between the subjective and physiological outcomes, despite longer testing durations, presents an argument that perhaps other, previously unexplored factors such as the influence of

Table 6
Pooled analysis: Centre of Pressure/weight shifts.

Outcome: Centre of Pressure weight shifts			
	Cham and Redfern, 2001	Lin et al., 2012a,b (Experiment 1)	Wiggermann and Keyserling, 2013
Study design	Cross-over trial	Cross-over trial	Cross-over trial
Setting	Laboratory	Laboratory	Laboratory
Age *Young = < 45 yrs;* *Old = > 45 yrs*	Young	Young	Young
Gender	M = 5 F = 5	M = 10 F = 3	M = 5 F = 5
Design	4 h standing	4 h standing	4 h standing
Breaks	2min walk every 30min	10 min at end of each hr sitting	10min walk/sit (half way through testing)
Interventions	6 mats (thickness): B = 7.1 mm C = 7.1 mm D = 11.2 mm E = 15.6 mm F = 16.9 mm G = 14.6 mm	Hard floor (force plate) + barefoot Hard floor + sports shoes Mat (12.5 mm thick) + barefoot Mat + sports shoes	4 mats: A (softest) B (softer) C (harder) D (hardest)
Footwear	Standardised hard sole shoes	Sport shoes	Standardised sport shoes (New Balance)
Control surface	Vinyl tile on steel plate (floor A)	Force plate	Linoleum tile on concrete
Testing times/ frequency	10min every 15-20min	5min each hr	10min each hr
Testing method	Using a force platform (Bertec Corp)	Using a force plate	Using in-shoe pressure sensors (0.13 mm thick F-Scan)
Definitions	Number of lateral weight shifts > 10% of total distance Sampling rate: 20 Hz	Number of lateral weight shifts from one leg to the other > 5% of total distance Sampling rate: 60 Hz	Weight shifting (whole body): Number of transitions between any of: a) > 80% body weight on L foot b) > 80% body weight on R foot c) ≥ 20% body weight each foot COP excursions (within foot): Av. Travel rate (cm/sec) on single foot when loaded min 20% body weight Medial-lateral (ML) Anterior-posterior (AP) Sampling rate: 20 Hz
Results	No significant differences among mats in 1st 3hrs Significant difference ($p = 0.04$) in 4th hr; with mat F (thickest, least stiff/soft, greatest work lost/energy absorbency) & A (control) the highest number of weight shifts Correlation b/w material properties of mats and weight shifts ($r = 0.5–0.7$) *Conclusion:* reduced weight shifts with increased elasticity, decreased energy absorption and increased stiffness.	No significant difference among the 4 intervention conditions (data not presented)	Significant difference between mats and weight shifting ($p < 0.01$) Increased number of weight shifts with time spent standing ($p = 0.01$) Hardest mat (D) – significantly more shifts than soft (A & B) and control Positive correlation with specific site discomfort ($p = 0.01$; $r = 0.24–0.35$) & overall discomfort ($p = 0.06$) Material properties of stiffness and work lost not related to weight shifting % of single foot stance increased with standing duration ($p = 0.03$) and correlated with discomfort ($p = 0.01$) Significant difference in COP excursion with softer mats producing less travel

psychological elements may play a role. One possibility is the influence of the Hawthorne effect. The Hawthorne effect refers to the changes in outcomes as a result of the attitudes and behaviours of the participants due to their awareness of being observed (Wickstrom and Bendix, 2000). The Hawthorne effect is a social concept sometimes compared to the placebo effect frequently described in pharmacology research. According to the theory behind the Hawthorne phenomenon, the awareness of standing on a mat or wearing cushioned insoles may be enough to trigger a positive response based on the common belief that softness is generally associated with comfort resulting in lower discomfort rating scores. An example of the potential presence of the Hawthorne effect is in the study by Wahlstrom et al. (2012). The flooring was replaced as a result of the involvement of the Swedish Work Environment Authority in order to address rising musculoskeletal complaints in the lower extremities of health care workers at the facility. The potential influence of the industrial context on attitudes of the workers towards this intervention, as recognised by the authors of this study, should be considered when examining the effect of the new floor on the level of musculoskeletal discomfort.

Another important consideration is the potential role of organisational and psychosocial hazards in the overall health and wellbeing at work and the development of work related MSDs. An Australian study by Oakman et al. (2014) conducted in three health care organisations in Victoria, demonstrated the relative contributions of physical, organisational and psychosocial factors in predicting work related musculoskeletal disorders (WMSDs). In particular, increasing age, female gender, higher levels of psychosocial hazards, high physical demands, low job satisfaction and poor work-life balance were significantly associated with discomfort. In the context of work performed whilst standing, it could be debated whether the degree of discomfort is a consequence of the particular flooring or shoe conditions or in fact the result of the interplay between the physical demand of low-level sustained muscle activity combined with the presence of psychosocial and organisational factors and cognitive task demands of the job. In a cross-sectional study of 407 standing assembly plant workers Werner et al. (2010) found that supervisor support was linked to a reduced risk of plantar fasciitis, whereas job dissatisfaction had the opposite effect. Similarly, a link was found between psychological distress and lower

Table 7
Pooled analysis: Leg volume.

Outcome: Leg volume				
	Cham and Redfern, 2001	Lin et al., 2012a,b (Experiment 1)	Lin et al., 2012a,b (Experiment 2)	Zander et al., 2004
Study design	Cross-over trial	Cross-over trial	Cross-over trial	Cross-over trial
Setting	Laboratory	Laboratory	Field	Field
Age *Young = < 45 yrs;* *Old = > 45 yrs*	Young	Young	Young	Old
Gender	M = 5 F = 5	M = 10 F = 3	M = 6 F = 8	M = 2 F = 11
Design	4 h standing	4 h standing	4 h standing	8 h shift
Breaks	2min walk every 30min	10 min at end of each hr sitting	No regular breaks	30min lunch & 2 × 10min breaks
Interventions	6 mats (thickness): B = 7.1 mm C = 7.1 mm D = 11.2 mm E = 15.6 mm F = 16.9 mm G = 14.6 mm	Hard floor (force plate) + barefoot Hard floor + sports shoes Mat (12.5 mm thick) + barefoot Mat + sports shoes	Hard floor (concrete) Mat (12.5 mm thick)	Mat (5/8 inch non-slip polyurethane) Insoles (Viscolas)
Footwear	Standardised hard sole shoes	Sport shoes	Not standardised sport shoes	Not standardised low heeled work shoes
Control surface	Vinyl tile on steel plate (floor A)	Force plate	Concrete	Wood block floor
Testing times/ frequency	Pre & post testing period	At commencement & at end of each hr	At commencement; half way through (2 h) & at end (4 h)	Pre & post shift
Testing method	Using a water tank displacement method	Measurement of thigh & shank circumference using Gulick tape at mid-shank (half way between knee & ankle) points; each measurement taken ×3 Food and fluid intake restricted	Measurement of thigh & shank circumference using Gulick tape at mid-shank (half way between knee & ankle) points; each measurement taken ×3	Measurement of R shank circumference in sitting using Gulick tape at 4 cm intervals from lateral malleolus up to 12 cm proximal to the lateral malleolus
Results	No statistically significant differences between flooring conditions Trend towards an increase in leg volume in all floor conditions with greatest mean difference for mat F (181cm3), A (179cm3) & D (178cm3); and least change for mat G (35cm3)	Shank circumference increased with standing in all conditions with greatest change in first 2 h Greatest change in shank circumference when standing on hard floor wearing sport shoes Fastest change in shank circumference when standing barefoot on hard floor Results support the use of mats in reducing leg swelling	Greater change in thigh and shank circumference when standing on hard floor compared to mat The increase in thigh and shank circumference was greater in field than in the laboratory experiment.	No statistically significant difference between mat, insoles and wood floor conditions ($p = 0.301$) An increase in leg volume in all 3 interventions ($p < 0.05$) Greatest increase for mat condition (Δ 29.75cm3); compared to insoles (Δ 27.37cm3) and wood floor (Δ 23.80cm3) Correlation between weight and leg volume change

leg, ankle and foot pain in the 1998 Quebec Health and Social survey (Messing et al., 2006) amongst standing workers. This evidence supports the possibility that psychosocial factors may be at least in part, involved in the development of discomfort in standing workers, however none of the included workplace studies in this review directly measured the impact of psychosocial factors on discomfort.

4.1. Study strengths and limitations

The design of the literature search was comprehensive and systematic in order to maximise the probability of identifying all relevant studies. The search was focused on literature from January 2000 to April 2016 to ensure most up to date information was considered. A rigorous approach to literature appraisal was employed in relation to study design, the assessment of the risk of bias, analysis of outcomes and reporting. The same three researchers were involved for consistency and assessment tools carefully selected from the literature to ensure quality of analysis.

However, this approach imposes some limitations. Firstly, in the absence of RCTs, only cross-over trials and pre and post design studies were considered. Studies with lesser levels of evidence such as cross-sectional studies were excluded. Further, the search was limited to the English language only and together these constraints may have prevented relevant studies from inclusion in the analysis. In order to maximise homogeneity, this review did not consider research involving walking or turning movements. This approach limits the applicability of findings to workplaces where work is characterised by constrained prolonged standing. The small number of trials, heterogeneity in study designs and significant biases impacted on the strength of evidence, and also precluded a meta-analysis. Due to the significant variations in study designs, some results obtained through grouping the outcomes by subjective and objective effects may need to be interpreted with caution. The generalizability of intervention effectiveness in real workplaces is also possibly limited due to a large proportion of results coming from laboratory studies (six studies in real workplaces compared to four studies in simulated workplaces).

5. Conclusions

Moderate level of evidence exists to support the use of cushioning materials for the reduction of perceived musculoskeletal discomfort of the lower limb and the lower back whilst standing at work. Unfortunately, conclusive assessment of the effectiveness of these interventions is not possible due to the limited number of studies, heterogeneity in study designs and study outcomes and the presence of numerous biases. Physiological changes were not consistently observed challenging the appropriateness and sensitivity of the tools currently available to detect changes in outcomes. There is not enough information in the existing literature to comment on the relative superiority of one type of cushioning material over another. There is only limited evidence that indicates insoles are superior to mats in reducing discomfort and fatigue in prolonged standing. There is insufficient information in the literature with only limited evidence to suggest that gender or age have no the effect on discomfort and fatigue. The possibility that previously uninvestigated psychosocial factors may also have a significant influence on musculoskeletal outcomes in standing workers is proposed for debate and further research. This review calls for larger, good quality prospective RCT intervention trials, designed to minimise the Hawthorne effect, and based in real work places that examine the influence of psychosocial and organisational factors on musculoskeletal discomfort whilst standing at work. These studies also need to be designed to take into account the differences between men and women and the younger and the older workers.

Conflicts of interest

We confirm that there is no conflict of interest regarding the material discussed in the manuscript.

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